

Transport Layer

Transport Layer: Provides process-to-process communication and multiplexing/demultiplexing using port numbers.

TCP

Transmission Control Protocol:
Connection-oriented, reliable,
byte-stream service.

UDP

User Datagram Protocol:
Connectionless, unreliable,
datagram service.

Sockets

Combination of IP address and
Port number, uniquely
identifying a process on a
machine.

TCP Connection Establishment:

3-Way Handshake: SYN -> SYN-ACK -> ACK.

TCP Congestion Control:

- Slow Start: Exponential increase of congestion window.
- Congestion Avoidance: Additive increase.
- Congestion Detection: Multiplicative decrease upon packet loss.

Well-Known Ports:

- HTTP: 80
- HTTPS: 443
- FTP: 20, 21
- SSH: 22
- DNS: 53